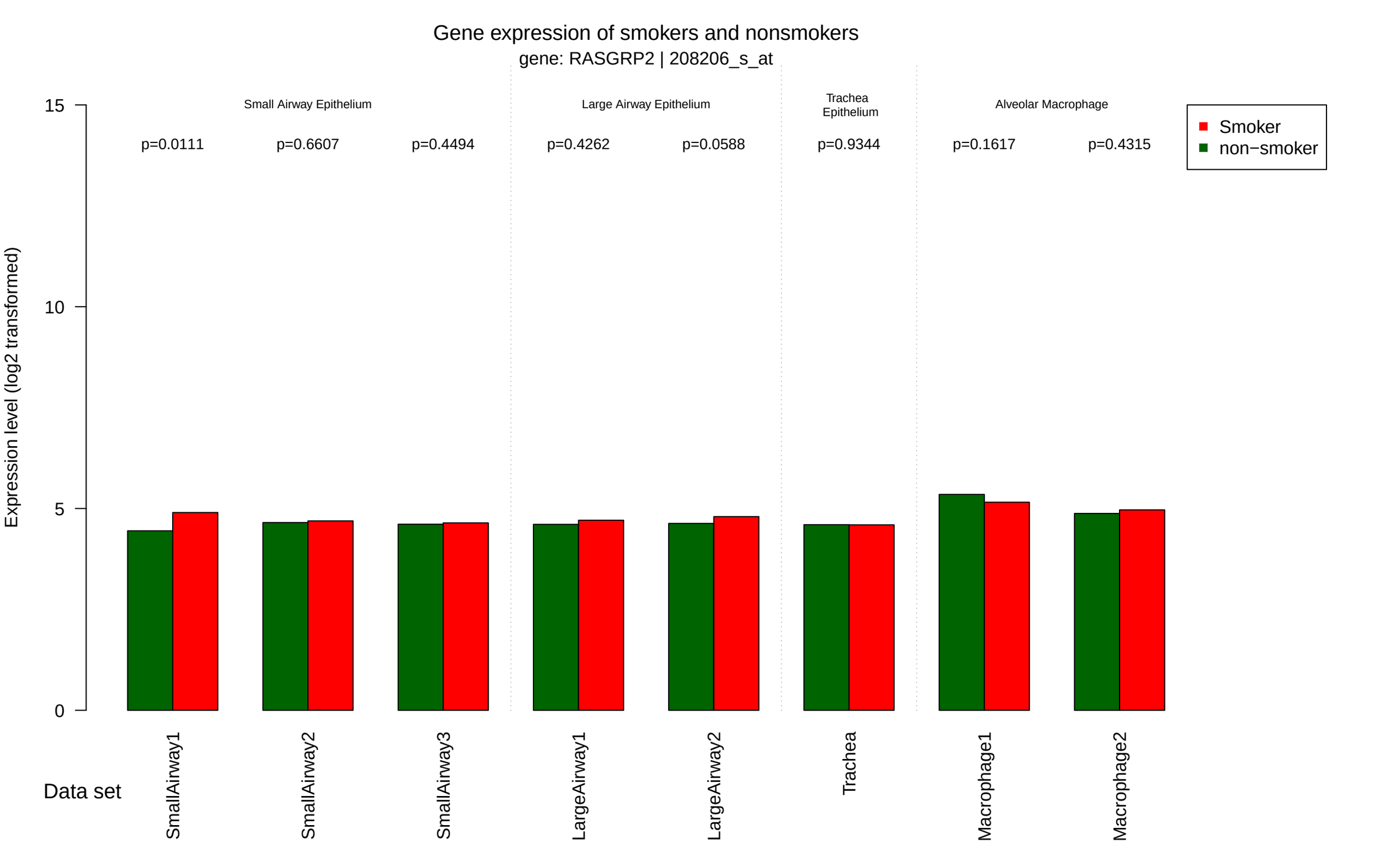
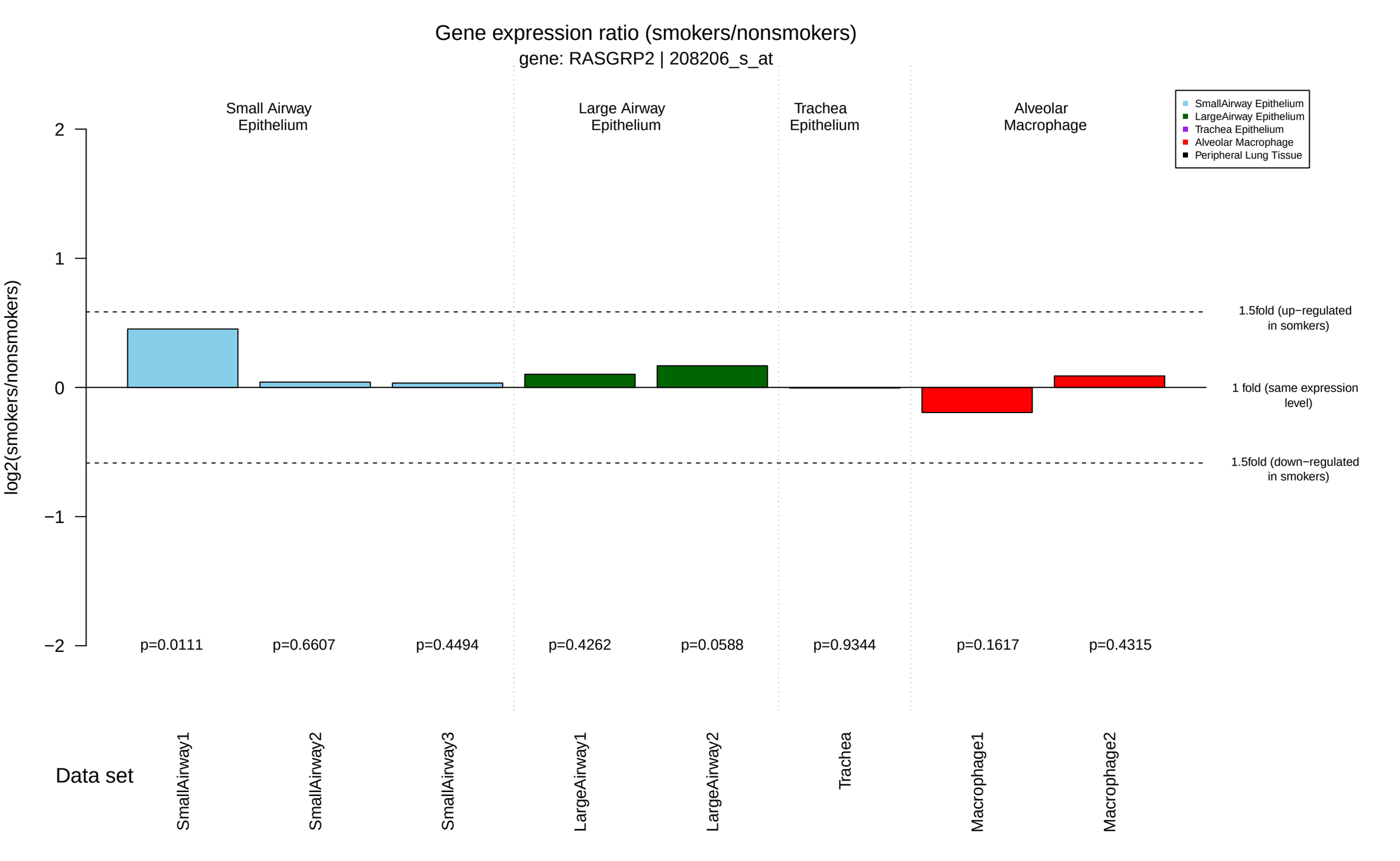
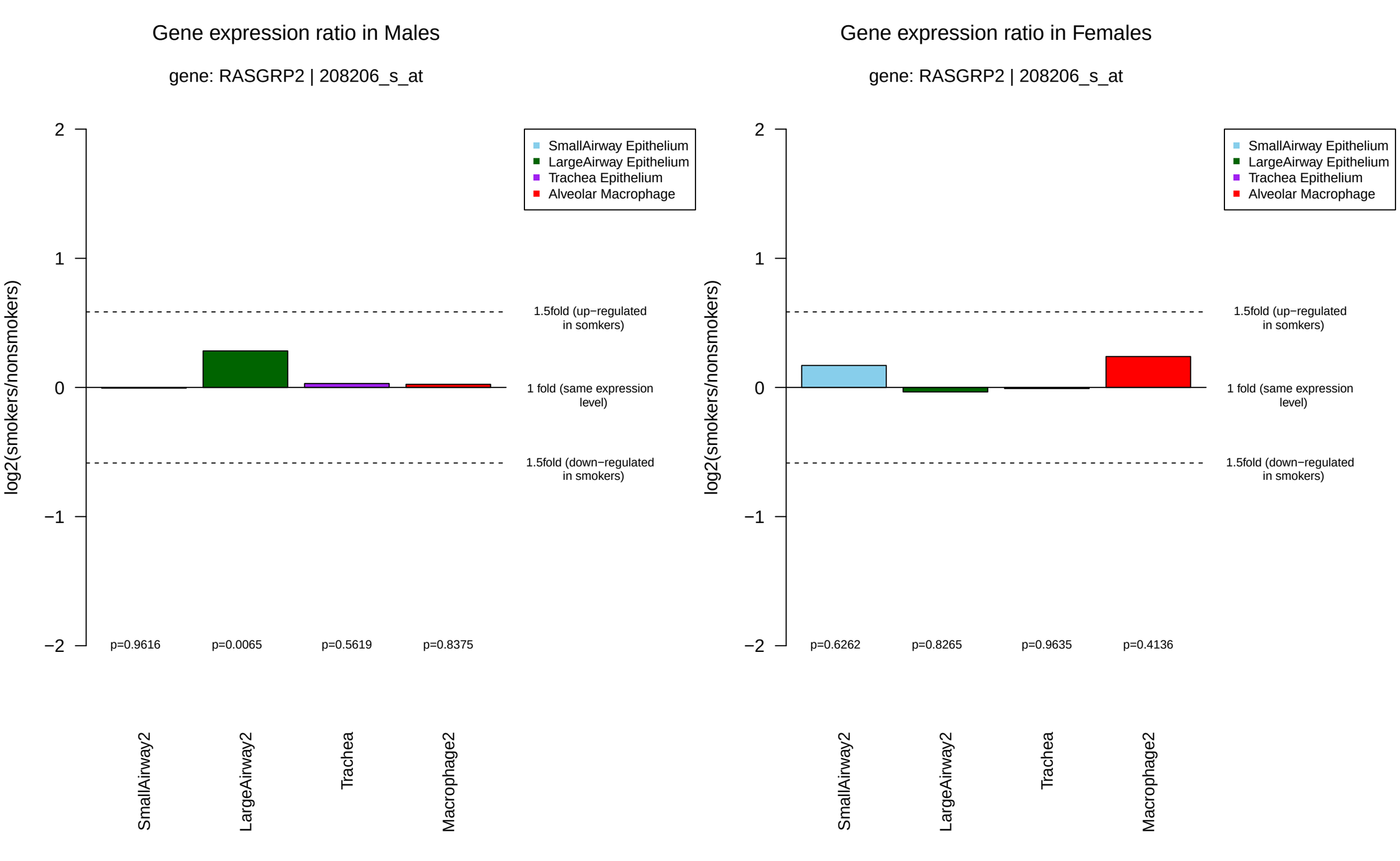
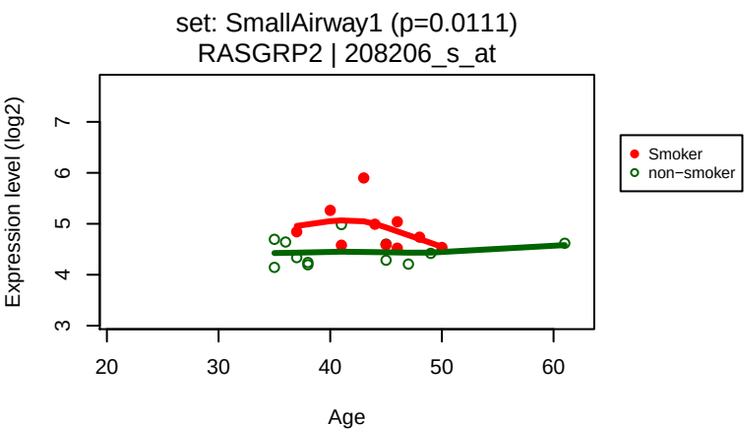
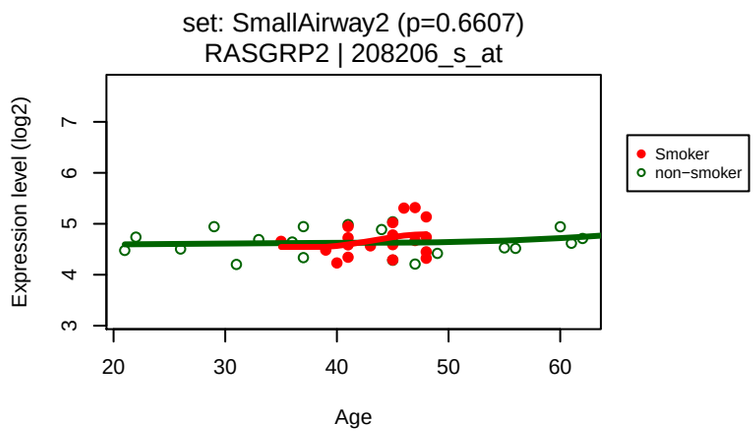
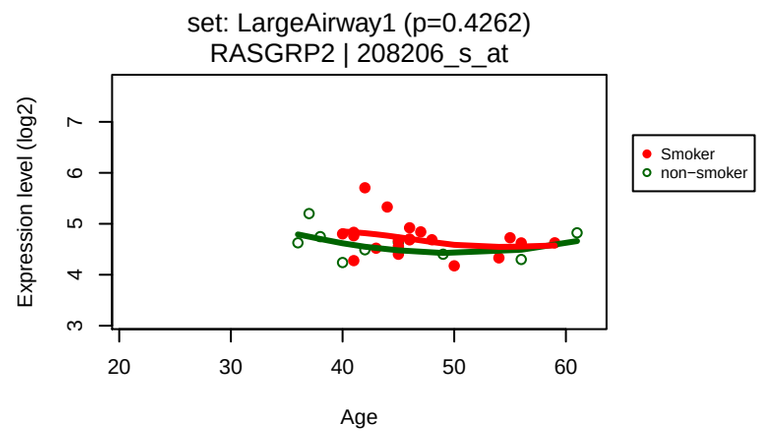

Gene expression of smokers and nonsmokers
gene: RASGRP2 | 208206_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0111
p=0.6607
p=0.4494
p=0.4262
p=0.0588
p=0.9344
p=0.1617
p=0.4315
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

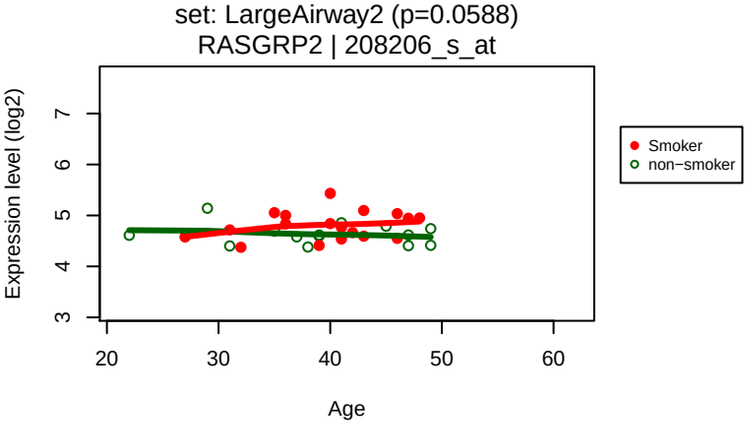

Gene expression ratio (smokers/nonsmokers)
gene: RASGRP2 | 208206_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0111
p=0.6607
p=0.4494
p=0.4262
p=0.0588
p=0.9344
p=0.1617
p=0.4315
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

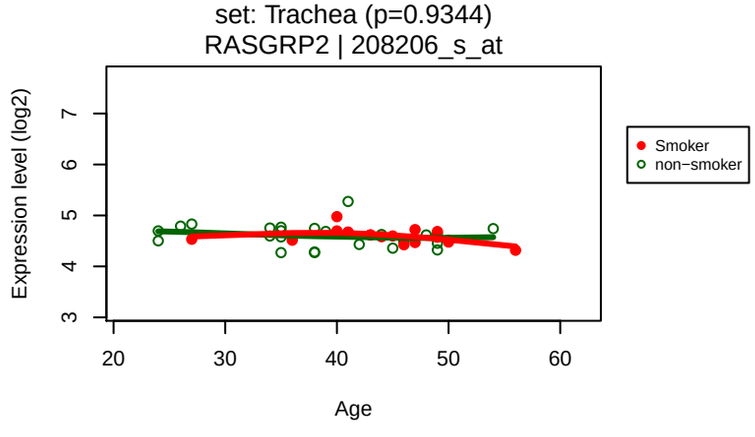

Gene expression ratio in Males
gene: RASGRP2 | 208206_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9616
p=0.0065
p=0.5619
p=0.8375
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RASGRP2 | 208206_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6262
p=0.8265
p=0.9635
p=0.4136
SmallAirway2
LargeAirway2
Trachea
Macrophage2

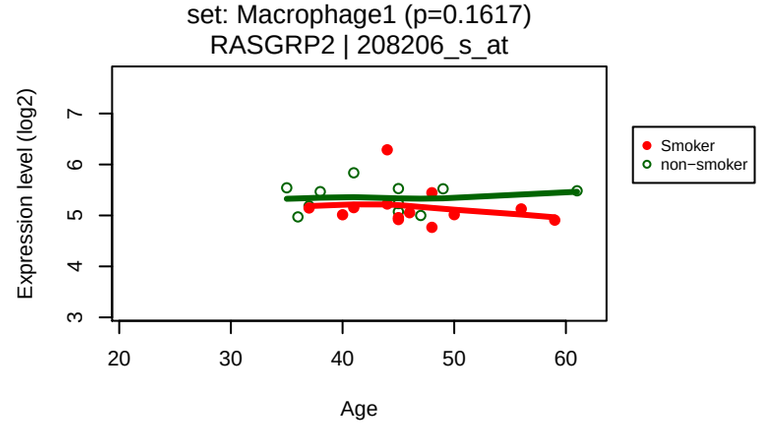

set: SmallAirway1 (p=0.0111)
RASGRP2 | 208206_s_at
Expression level (log2)
Age
Smoker
non−smoker

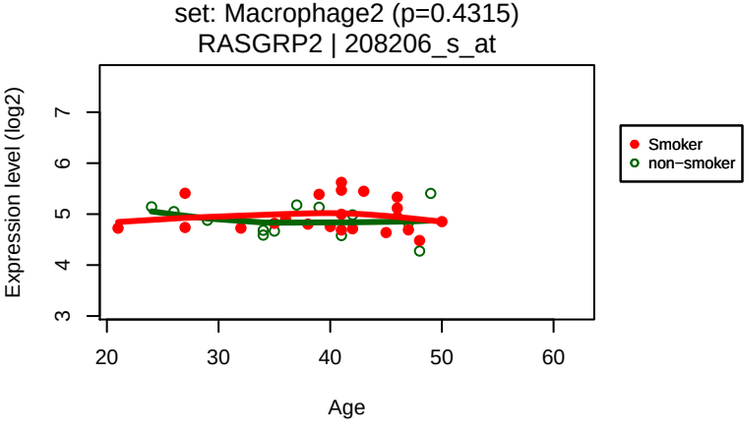

set: SmallAirway2 (p=0.6607)
RASGRP2 | 208206_s_at
Expression level (log2)
Age
Smoker
non−smoker

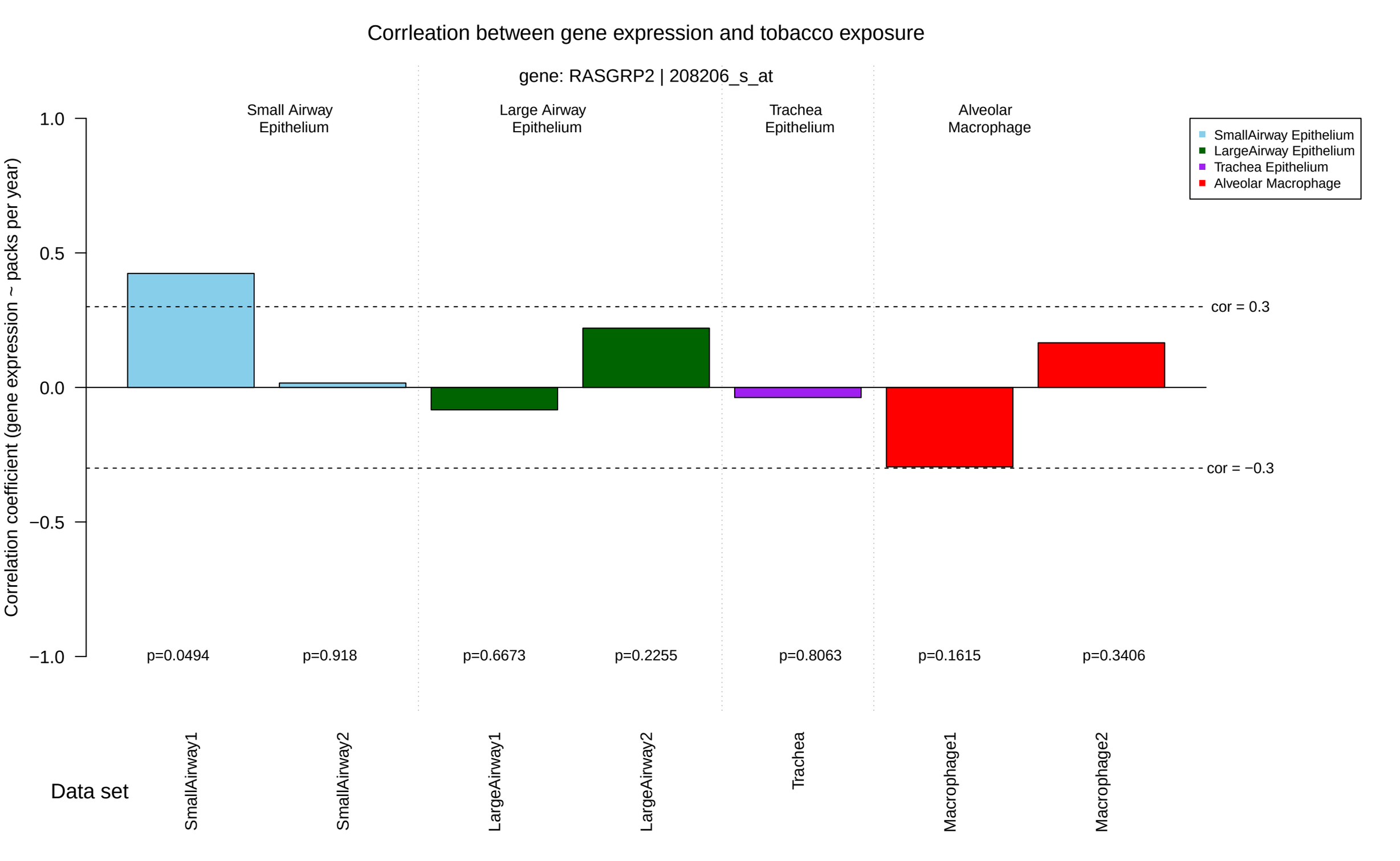

set: LargeAirway1 (p=0.4262)
RASGRP2 | 208206_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0588)
RASGRP2 | 208206_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9344)
RASGRP2 | 208206_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1617)
RASGRP2 | 208206_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4315)
RASGRP2 | 208206_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RASGRP2 | 208206_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0494
p=0.918
p=0.6673
p=0.2255
p=0.8063
p=0.1615
p=0.3406
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2